

Applications of Mathematics (Maths Interview Preparation for International students) Syllabus

Fall Semester, 2025

Department	Math&Comp.Sci.	Subject	Appl. of Maths	Credits	3	Class Hours/Wk	3
Instructor		Office	E-mail		Office Hour		
Jerome Tambour		3507(T148)	jerome@ksa.hs.kr		Mon 3,4, Fri 1,2		

Google Classroom: 6jjwq65q

1. Course Description

Course Objectives	
Applications of Mathematics – Maths Interview Preparation is designed for senior international students to prepare math interviews for university admission, in particular KAIST math interview for international admission. To achieve that goal, the course emphasizes on two aspects: - the review of the material from the PreCalculus and Calculus courses. To fit the range of most interviews, we will use the curriculum of AP Calculus tests. - training on the specific aspects of an interview: interactivity, time constraints, material constraints,...	
Contents	
Content: - Real numbers - Functions - Counting problems - Polynomial functions and complex numbers - Trigonometry - Exponential and logarithms - Limits of sequences and functions - Continuity - Derivatives - Applications of derivatives	Note: the course being designed to prepare students for the math interviews of the universities they applied for, the exact topic reviewed during the course will be decided with the students accordingly.

2. Textbook & References

PreCalculus, (Metric Version), James Stewart, 8th edition.

Calculus (Metric Version), James Stewart, 9th edition.

3. Core Target Competencies

Problem solving: solving problems related to fundamental topics of mathematics. Reasoning: proving and disproving statements; using correct rules of logic. Communication: write a mathematical reasoning in a clear and rigorous way, present one's work. Attitudes and practices: learning course material in an organized way. Information ethics: use critical thinking to analyze claims.

4. Grading

Homework & Activities	Quizzes	Interview	Final Exam	Attendance	Total
20%	20%	10%	40%	10%	100%
40	40	20	80	20	200 points

1) Homework and Activities: If you don't submit your homework on time, some points will be reduced from the maximum score. The homework for a week are due on the first class of the following week (unless stated otherwise). Part of the Activities grade is based on student presentations of the solutions of given problems.

2) Quizzes: Quiz at the beginning of the first class of each week on the topic of the previous week.

3) Interview: Two interviews in the style of KAIST math interviews (face-to-face, no document, 10 to 15 minutes, one or two problems to solve) will be held. The best result out of the two interviews will be kept as the grade.

4) Examinations: Final exam during Week 17 on the topics studied in class. No calculator or electronic device, and no document whatsoever. No midterm examination. If the final examination is missed due to official leave or medical leave, a re-evaluation will be conducted.

5) Attendance: absence: -3 points except illness or official cases (with confirmation from our school infirmary in case of illness and contact to teacher in advance in case of official absence)

tardiness : -1 point

5. Tentative schedule:

	Session 1	Session 2	Session 3
	Monday 8 nd period	Friday 3 rd period	Friday 4 th period
Week 1	Intro + Real numbers	Real numbers	
Week 2	<i>Quiz Real numbers</i> + Functions	Functions	
Week 3	<i>Quiz Functions</i> + Counting problems	Counting numbers	
Week 4	<i>Quiz Counting problems</i> + Polynomials	Polynomials	
Week 5	<i>Quiz Polynomials</i> + Trigonometry	Trigonometry	
Week 6	Trigonometry	<i>Quiz Trigonometry</i> + Exp and log	
Week 7	Exp and log	개천절	
Week 8	Chuseok break		
Week 9	Midterm week (no examination, no class)		
Week 10	<i>Quiz Exp and log</i> + Limits	Limits	
Week 11	Limits	SAC	
Week 12	<i>Quiz Limits</i> + Continuity	Continuity	
Week 13	<i>Quiz Continuity</i> + Derivatives	Derivatives	
Week 14	Interview 1		
Week 15	<i>Quiz Derivatives</i> + Appl. of derivatives	Appl. of derivatives	
Week 16	Interview 2		
Week 17	Final examination		

Note: This syllabus may be updated depending on school circumstances.